

Contents

Types of Facts Tables:	34
Fully additive measure/ fact:	36
Semi-additive	37
Non additive measures	38
Derived measures/facts :	40
Factless Fcat Table	41
Textual Measures:	42
Surrogate key:	43
Features	58
Major Advantages	58
Cubes	58
Peeling back 3 layers of BI and Analytics software	59
<i>Data Lake architecture</i>	63

Top-down Approach

- Proposed By Dr. Bill Inmon
- DWH is created first and then the Data Marts are derived from it.
- Data marts have consistent data and Subject Oriented Data.
- It is robust against business changes. The cost of the project is very high.

Refer L [Bhaskar Jogi](#) youtube channel for DW

Refer: https://medium.com/@arjunagarwal_22593/building-a-data-warehouse-79923addab0c

https://www.youtube.com/watch?v=llmtOVc8C1E&list=PLskvRY5fTxgcW3m_hDhqmW0QJftoahL12&index=3

<https://www.youtube.com/watch?v=Ddzn9siwx6Q&list=PLskvRY5fTxgf36tOnq9P82IGcu29EwpXm&index=5>

[edureka:](#)

https://www.youtube.com/watch?v=J326LIUrZM8&list=PL9ooVrP1hQOEDSc5QEBI8WYVV_EbWKJwX

[Erwin:](#)

<https://www.youtube.com/watch?v=IJ-5vC3pjHA>

https://www.youtube.com/watch?v=YwVT_pdxbkE

https://www.youtube.com/watch?v=YwVT_pdxbkE

<https://www.youtube.com/watch?v=0Vby5YHVkYY>

https://www.youtube.com/watch?v=m4S4Mo8U_ao&list=PL2-GO-f-XvjBnTL9CDPBXVTDO_UT_k28O&index=11

dimensional modeling interview questions:

<https://tekslate.com/interview-questions-on-dimensional-data-modeling>

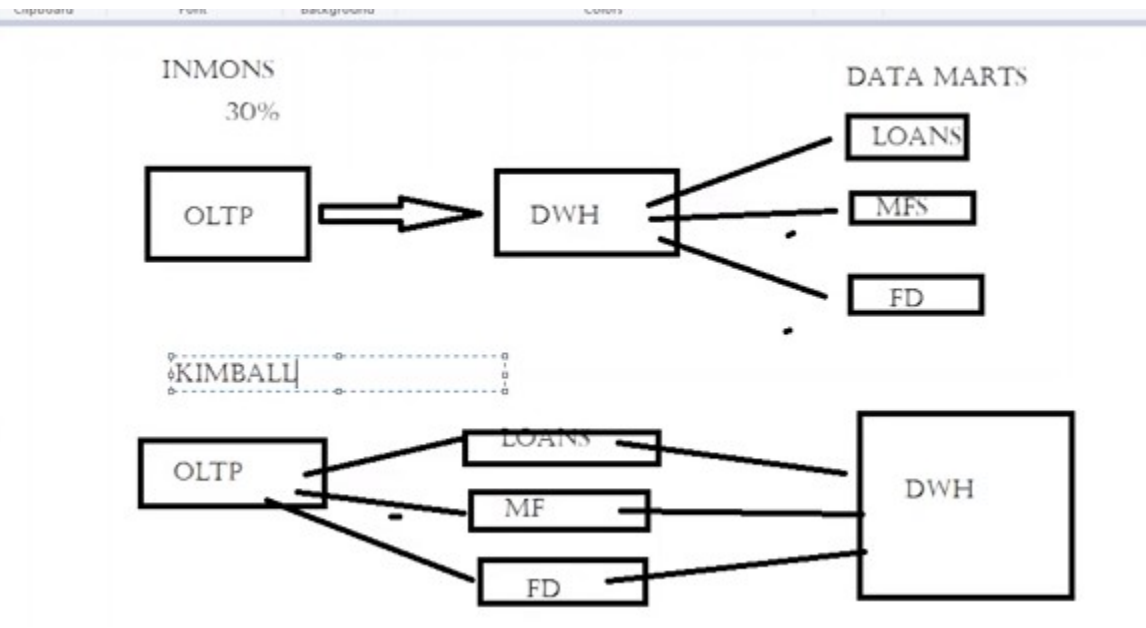
build datware house with Erwin in 4 steps

https://www.youtube.com/watch?v=YwVT_pdxbkE&t=5s

Bottom-up Approach

- Proposed By Dr. Ralph Kimball
- Data marts are created first and then integrated and a comprehensive DWH is created.
- As the data marts are created first, business solutions can be answered quickly.

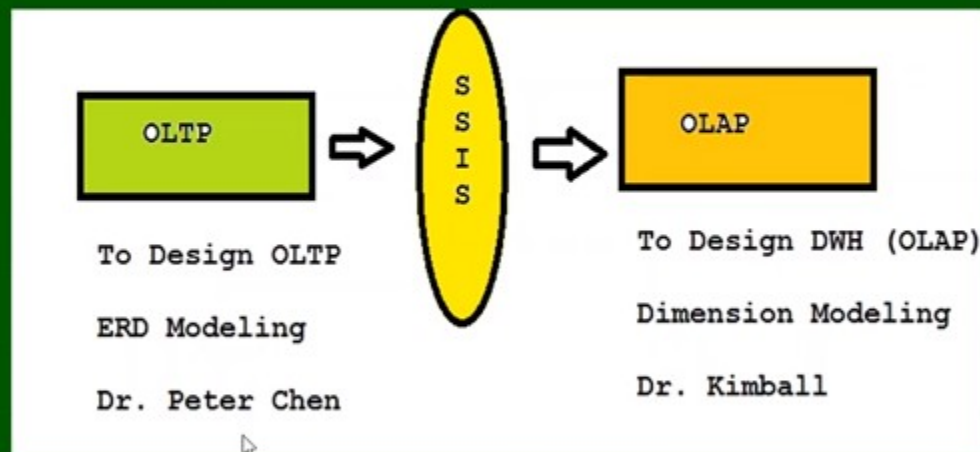
Activate Windows
Go to Settings to activate Windows.



Agenda

- Design Concepts of DWH
- Intro to Dimension Modeling
- What is a Fact or Measure
- What is a Dimension
- Dimension Hierarchies
- Types of Dimension and Facts
- Intro to Schemas
 - Star Schema
 - Snow Flake Schema

Design Concepts – OLTP and OLAP (DWH)



Activate Windows
Go to Settings to activate Windows.

OLTP to OLAP Table Conversion

- OLTP Contains
 - Master Tables
 - Product
 - Course
 - Student etc
 - Transactional Tables
 - Payment Details
 - Order Details etc
- In OLAP
 - Master tables converted as Dimension Tables
 - Transactional tables converted as Fact Tables

In OLTP master n transactional tables will be there

In OLAP → master table converted as dimension tables

→ Transaction tables are converted as fact tables

Master (pk,string,small data)

```

-----
                pk
ProuctMaster (pid, name, price)    50 gb
                pk
CustomerMaster (cid,name,addr,city,phone)  40gb

```

Transactional (fk,numeric,large data) - 500gb

```

-----
        fk    fk
Sales    (pid,cid,dos,qty,amt)

```

OLTP ----- DWH

```

-----
master ----- Dimension
-----

```

```

ProductMaster  ----- DimProduct
CustomerMaster ----- DimCustomer
Sales          ----- FactSales

```

Dimension tables are loaded using incremental loading

Scd(slowly changing dimensions these methods for dimension tables not for facts tables)

For fact tables we can't use incremental loading

For fact table data load will be full load OR INCREMENTAL LOAD

Types of Dimensions

- Role Playing Dimension
- Conformed Dimension
- Junk Dimension
- Degenerate Dimension
- Slowly Changing Dimension
- Rapidly Changing Dimension

Activate Windows
Go to Settings to activate Windows.

Types of Measures/Facts

- Fully Additive Measures/Facts
 - Can be added across all the associated dimensions
 - Eg: Sales Amount
- Semi-Additive Measures/Facts
 - Can be added across only few dimensions rather than all dimensions.
 - Eg: Customer Balance at the Q1 and Q2

Activate Windows
Go to Settings to activate Windows.

Fact Less Fact Table

www.GoOnlineTrainings.com

Fact Less Fact Table

- Fact Table which does not contain any measures or facts
- Fact Table contains only key Attributes

Fact Less Fact Table	
DimStudent	
PK	SID
	Name
	Gender

FatctAttendance	
Fk	SID
Fk	CID
Fk	TimeKEY

DimCourse	
PK	CID
	Name
	Fee
	Type

DimTime	
PK	TimeKEY
	Year
	Qtr
	Month
	Week
	Day

Activate Windows
Go to Settings to activate Windows.

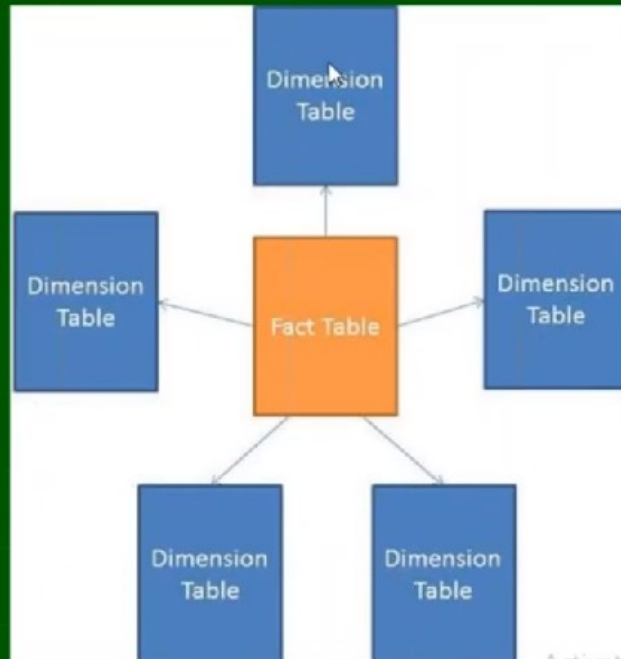
In the screen factAttendance is fact less fact table bcz it don't have any measures or facts

How Fact and Dimension tables are organized in a DWH/Cube?

- Star Schema
- Snow Flake Schema

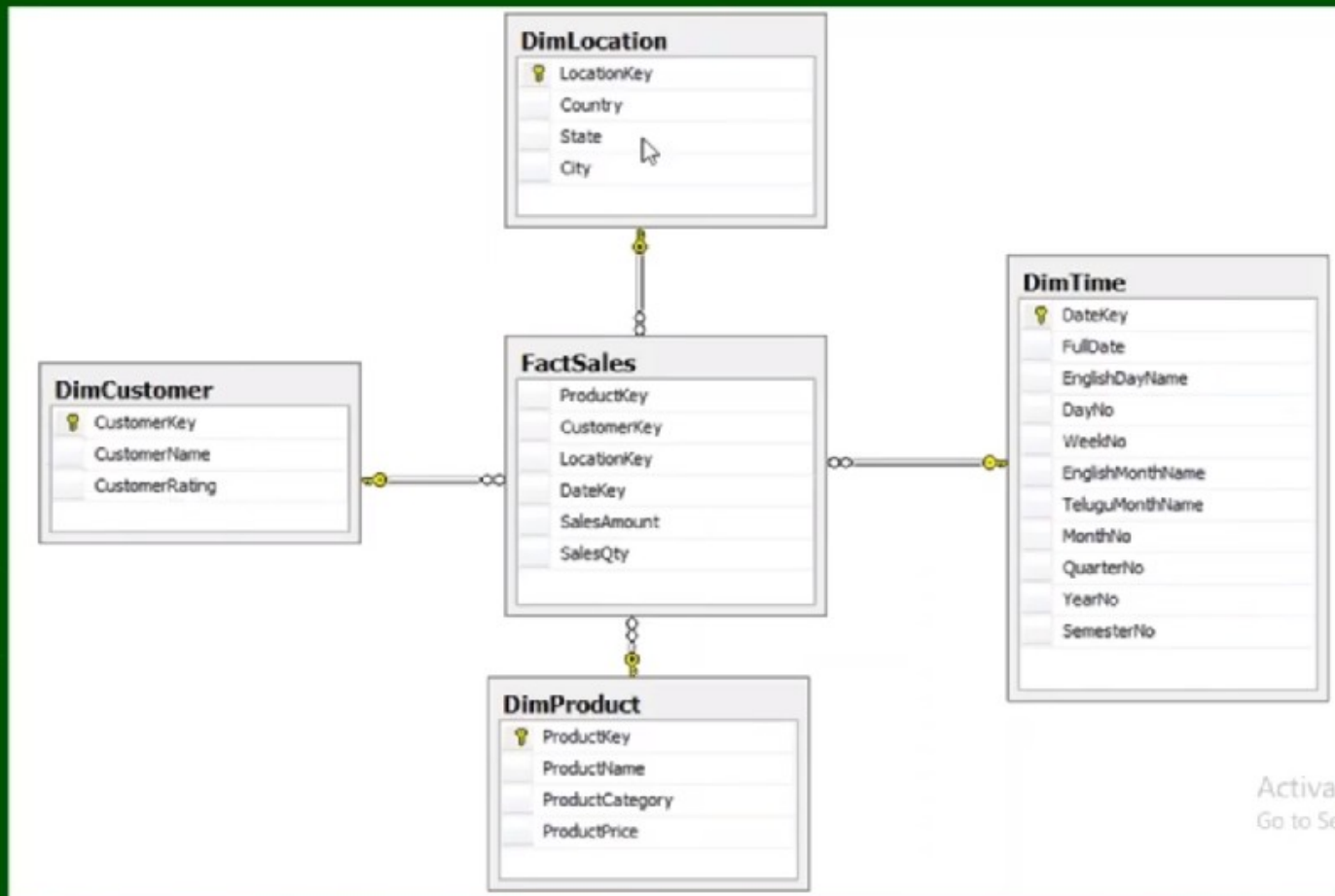
Star Schema

Star Schema means
Dimension Tables are
directly linked to the
Fact Table



Activate Windows
Go to Settings to activate Windows.

Star Schema



Activate Win
Go to Settings to

In above image in table Factsales if sales amount , salesqty column are not there then it will become fact less fact table.

Snow Flake Schema

Dimension Tables are indirectly linked to the Fact Table

Or

Dimension table is linking to another dimension table



Activate Windows
Go to Settings to activate Windows.

OLTP vs OLAP

OLTP					
emp			dept		
eid	name	deptid	deptid	name	
101	John	2	1	Sales	
102	Ram	2	2	Tech	
103	Kiran	1			
104	David	2			

OLAP			
	EID	NMAE	DNAME
	101	John	Tech
	102	Ram	Tech
	103	Kiran	Sales
	104	David	Tech

Activate Wi
Go to Settings t

OLTP

Writing

Split Tables

Normalization

-ve

More tables

More Joins

More time to read the data (Slow in Read)

OLAP (DWH)

Reading

Combine Table

Denormalization

Less tables

Less Joins

--ve

Slow in wrting of the data|

Star Schema Vs Snow Flake Schema

Star Schema	Snow Flake Schema
1.Has redundant data and hence tough to maintain /change	1.No redundancy and hence more easy to maintain /change
2.Less complex queries and hence easy to understand	2.More complex queries and hence less easy is to understand
3.Less foreign keys so faster execution time Of cube	3.More foreign keys so takes more execution time of cube
4.Has De-Normalized tables	4.Has Normalized tables
5.Good to use for large databases	5.Good to use for small databases
6.Less number of joins	6.More number of joins
7.When dimension table has less number of rows we can go for this	7.When dimension table is relatively big in size it is better as it reduces space

Activate V
Go to Setting

Most of the traditional DWH will be snowflake schema

Star Vs. Snow Flake Schema

	ETL	Cube Process
Star Schema	Slow	Fast
Snow Flake Schema	Fast	Slow

Note: Star Schema is recommended for Data Warehouse

ETL Mapping Document

- To map the OLTP columns with DWH columns and perform business rules during ETL process, while creating DWH database.

Data Warehouse Keywords

- OLTP vs OLAP
- Data Mart vs Data Warehouse
- ODS (Operational Data Source)
- CUBE
- UDM (Unified Dimension Model)
- Dimension
- Measure or Fact
- Measure Group
- Hierarchy
- Member
- Attribute
- Schemas
 - Star
 - Snowflake

Activate Windows
Go to Settings to activate Windows.

To Design DWH System

- 1) OLTP System
- 2) Report Requirements/Data Analytics/Data science

OLTP

ProdM

CatM

CustM

Sales

DWH(OLAP)

- 1) Reduce no of tables in DWH

Activate Windows
Go to Settings to activate Windows.



Type here to search



8:55 AM
5/23/2018

Amazon OLTP.xlsx - Microsoft Excel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
14							1	Vamsi	10							
15			SALECT	Slow			2	Bhaskar	10							
16			Normalization				3	Balu	20							
17							4	Kiran	30							
18							5	John	10							
19							6	David	30							
20			OLAP													
21					Emp		EID	Name	DeptName							
22			Historical data				1	Vamsi	Sales							
23			Large Data				2	Bhaskar	Sales							
24			Read Only				3	Balu	Accs							
25			De-Normaliztion				4	Kiran	Marketing							
26							5	John	Accs							
27			SELECT	Faster												
28																
29			INSERT	Slow												
30			UPDATE	Slow												
31			DELETE	Slow												
32																
33																
34																
35																
36																
37																
38																

Activate Windows
Go to Settings to activate Windows.

MappingSheet-DimCustomer DimTime OLTP DWH Sheet1 OLTP Trucks OLAP Trucks Sheet2

Ready Type here to search

8:55 5/23/2018

More indexes required in OLAP then reading query will be fast.

Where as in OLTP less indexex are required bcz as indexes are more then writing will be slowdown.

More indexes then DML operation will be slow .DML operations are more in OLTP,

DML operations are less in OLAP. We will use select clause more in OLAP

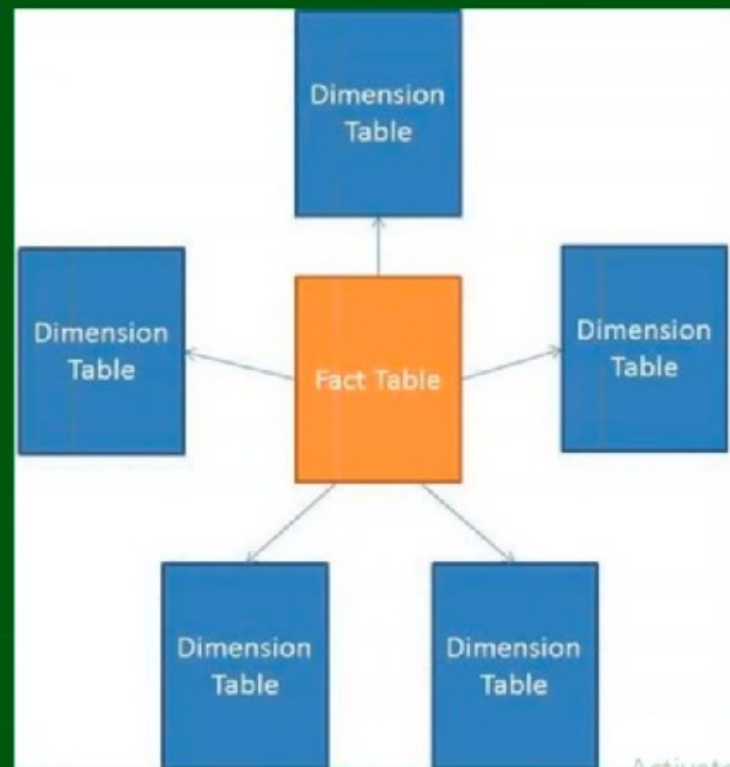
How Fact and Dimension tables are organized in a DWH?

Press Esc to exit full screen

- Star Schema
- Snow Flake Schema
- Galaxy (Double Star Schema)

Star Schema

Star Schema means
Dimension Tables are
directly linked to the
Fact Table

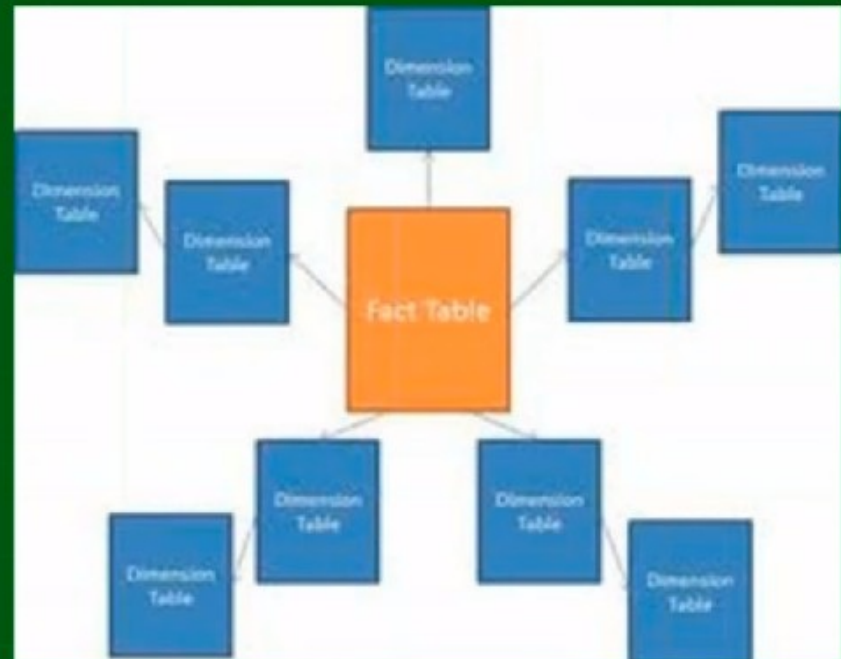


Snow Flake Schema

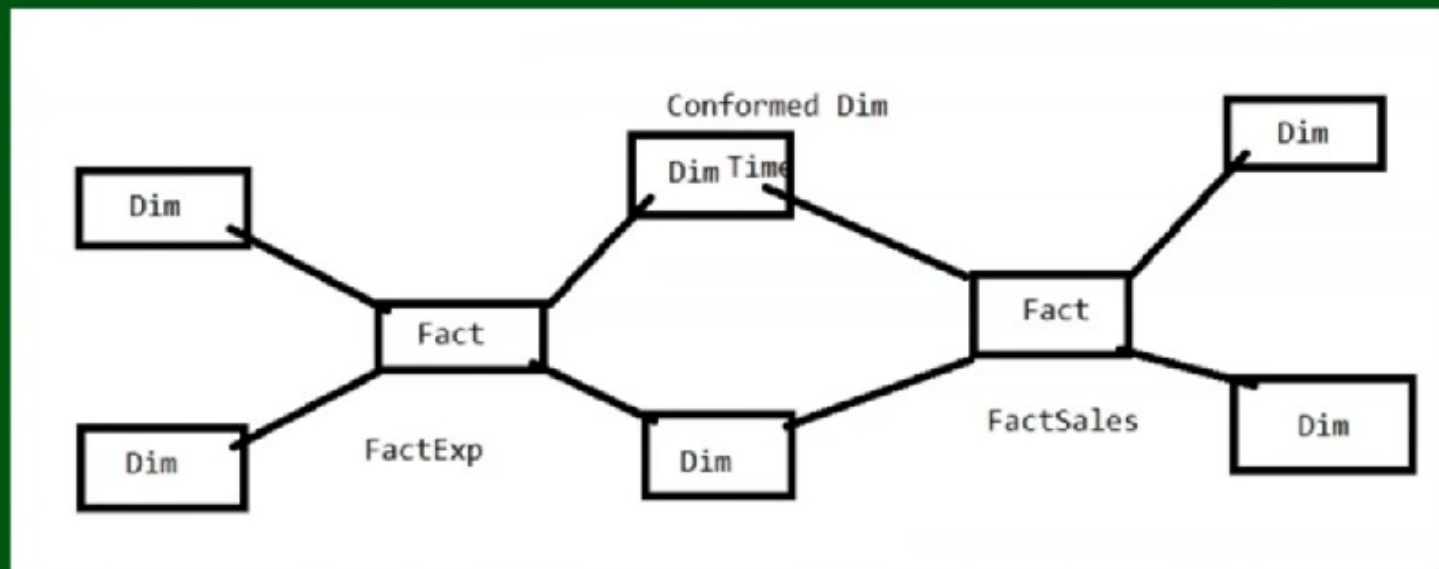
Dimension Tables are indirectly linked to the Fact Table

Or

Dimension table is linking to another dimension table



Double Star/Galaxy Schema



DWH Best Practices

Less No Of Tables

Study fully OLTP DB

Study Reporting Requirements/ Data Analytics

Proper Relationships b/w tables (Dim n Fact)

Star is recommended as speed is high.

In real time snowflake schema will be there

Types of Dimensions

- Role Playing Dimension
- Conformed Dimension
- Junk Dimension
- Degenerate Dimension
- Slowly Changing Dimension
- Rapidly Changing Dimension
- Inferred Dimension

Conformed Dimensions

Conformed Dimension

- A conformed dimension relates to multiple fact tables within the same DWH.
- **DimDate** is a common conformed dimension because its attributes (day, week, month, quarter, year, etc.) have the same meaning when joined to any fact table.

Activate Windows
Go to Settings to activate Windows.

Conformed dimensions are those dimensions which have been designed in such a way that the dimension can be used across many fact tables in different subject areas of the warehouse. It is imperative that the designer plan for these dimensions as they will provide reporting consistency across subject areas and reduce the development costs of those subject areas via reuse of existing dimensions. The date dimension is an excellent example of a conformed dimension. Most warehouses only have a single date dimension used throughout the warehouse.

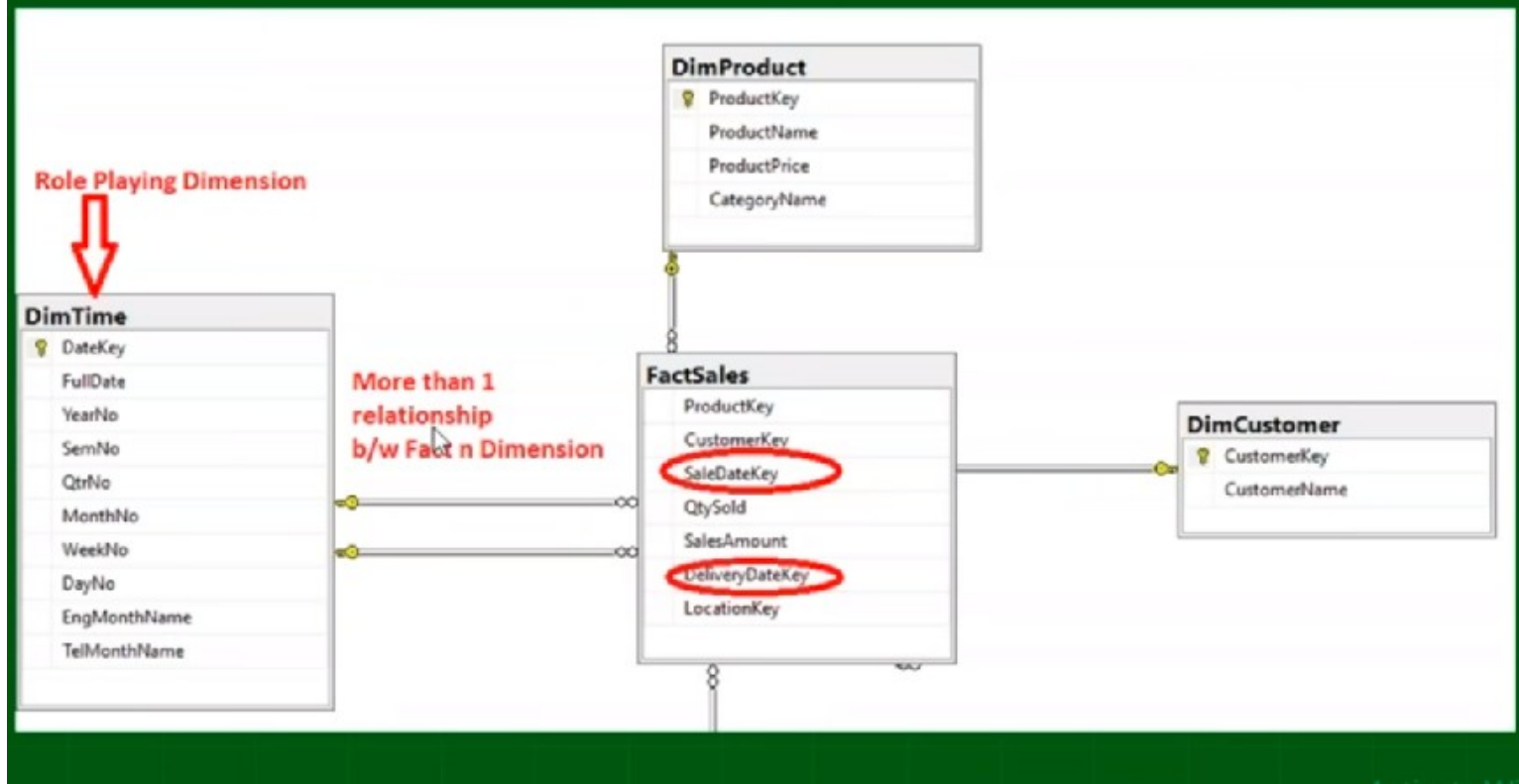
Confirmed dimension will be available in galaxy /double stat schema

Conformed Dimension- This is used in multiple locations. It helps in creating consistency so that the same can be maintained across the fact tables. Different tables can use the table across the fact table and it can help in creating different reports.

Role playing dimension:

More than 1 relationship b/w fact and dimension table. In below image dimTime Table has 2 relation ships with fact sales table side sales date ky, delivery datakey

Role Playing Dimension



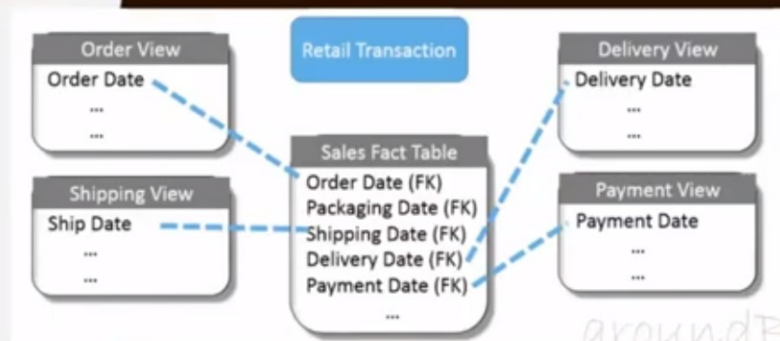
Below screnn shot is from this video :aroundbi channel

<https://www.youtube.com/watch?v=gynrzgH6c78>

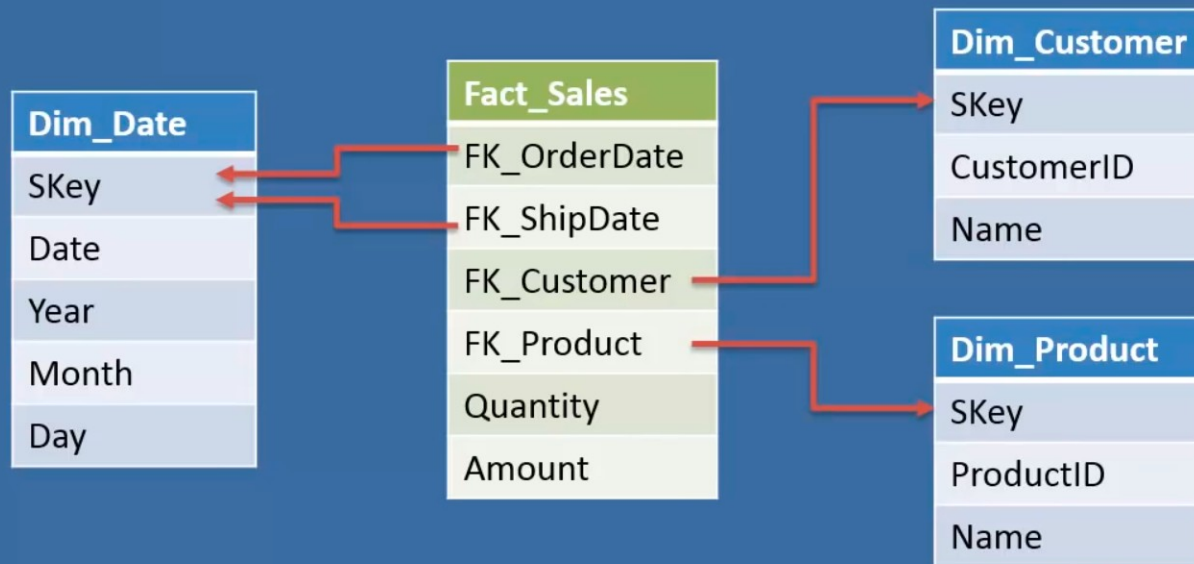
Role Playing Dimension

Highlights

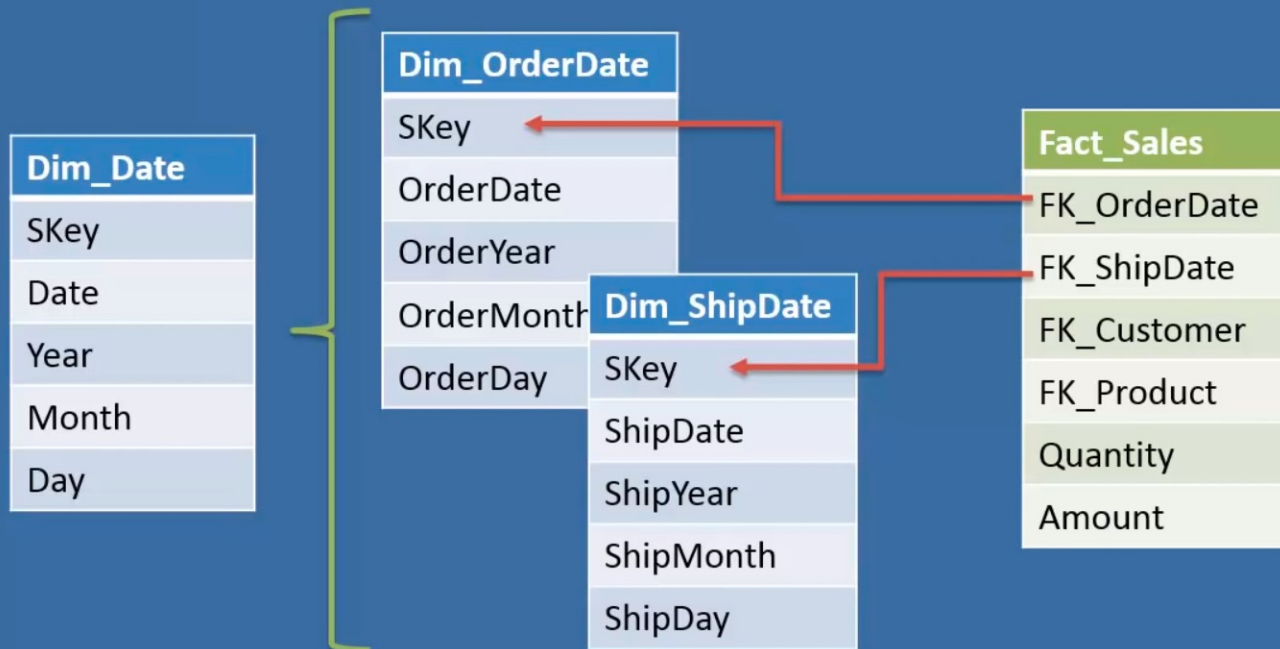
- Role playing dimensions - appear several times in the same fact table with unique labels
- DWH structure - create independent views for each fact table foreign key



Role Playing Dimension



Role Playing Dimension



Junk Dimension

- Junk dimensions are used to reduce the number of dimensions in the dimensional model
- and reduce the number of columns in the fact table
- It will help us to reduce size of the fact table

Transactional business processes typically produce a number of miscellaneous, low-cardinality flags and indicators. Rather than making separate dimensions for each flag and attribute, you can create a single *junk dimension* combining them together. This dimension, frequently labeled as a *transaction profile dimension* in a schema, does not need to be the Cartesian product of all the attributes' possible values, but should only contain the combination of values that actually occur in the source data.

Remove unnecessary details like payment details from fact table and create a different table that is called junk dimension

FactSales	FK	FK	FK			FK	FK	FK
	CustomerKey	ProductKey	SaleDateKey	QtySold	SalesAmt	DeliveryDateKey	LocationKey	DimSalesDetails
	102	2	1	1000	20000	3	6	5
	103	4	3	5	1750	5	1	
	101	6	3	5	125			
Junk	Pk							
DimSalesDetails	SalesDetailID	Mode	Currency					
	1	Cash	INR					
	2	CASH	USD					
	3	CARD	INR					
	4	CARD	USD					
	5	NetBanking	INR					
	6	NetBanking	USD				act	

Degenerate Dimension

- The term "**degenerate dimension**" was originated by Ralph Kimball.
- **Degenerate dimension** describe a dimension field in the fact table that doesn't have a corresponding dimension table.
- Placing these text attributes will slowdown the performance of Fact Tables
- Eg: Voucher Number, Bill Number etc

FactSales	FK	FK	FK			FK	FK	FK	
	CustomerKey	ProductKey	SaleDateKey	QtySold	SalesAmt	DeliveryDateKey	LocationKey	SalesDetailsKey	Bill No
	102	2	1	1000	20000	3	6	5	A-123
	103	4	3	5	1750	5	1	1	A-849
	101	6	3	5	125				

Slowly Changing Dimension

- In Data Warehouse there is a need to track changes in dimension attributes in order to report historical data.
- Slowly changing dimensions are used when we wish to capture the changing data within the dimension over time

Historical data will be maintained SCD i.e SCD1, SCD2, SCD3

Inferred Dimension

- Inferred Dimension are called as Late Arriving Dimensions
- Early arriving Facts, Late arriving Dimensions

Below fact came first ,dimension came late so we will follow 2 steps

(1) Either we will drop FK on fact table

Or

We will update key val in dimension table with other values as null later we will update remaining columns

Summery

- Confirmed is some thing which can be shared by multiple Fact Tables or multiple Data Marts.
- Junk Dimensions is grouping flagged values
- Degenerate Dimension is something dimensional in nature but exist fact table.

DWH Best Practices

- Reduce No of Dimensions
- Reduce No of columns in Fact Table
- Reduce size of Keys (PK and FKs)
- Have Junk Dimensions
- Have Degenerate Dimensions

Types of Facts Tables:

Types of Fact Tables

- Fully Additive Measures/Facts
- Semi Additive Measures/Facts
- Non Additive Measures/Facts
- Derived or Calculated Measures/Facts
- Fact less Measure/Facts
- Textual Measures/Facts

Types of Measures/Facts

- Fully Additive Measures/Facts
 - Can be added across all the associated dimensions
 - Eg: Sales Amount
- Semi-Additive Measures/Facts
 - Can be added across only few dimensions rather than all dimensions.
 - Eg: Customer Balance at the Q1 and Q2

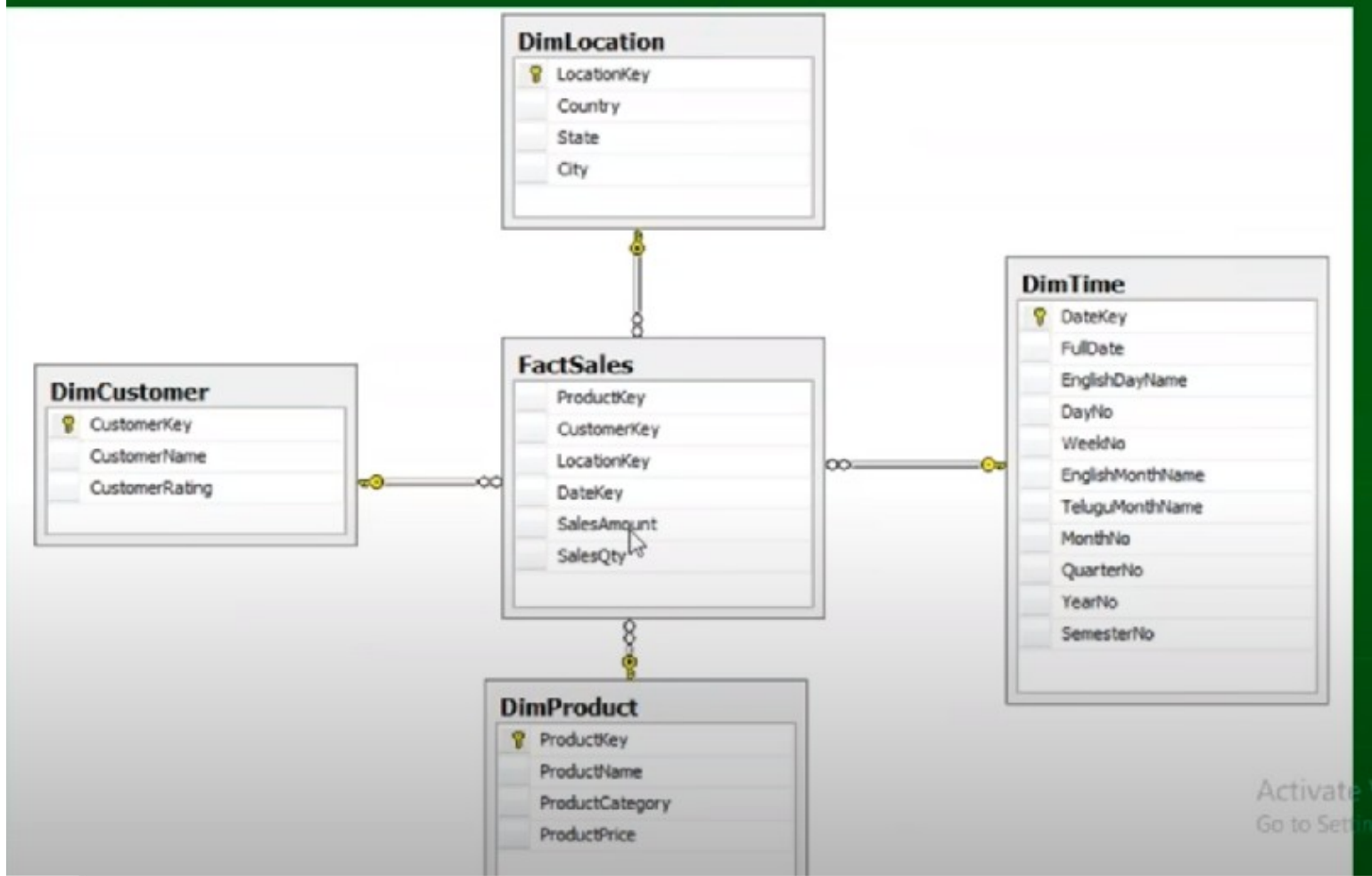
Activate
Go to Settings

Fully additive measure/ fact:

The numeric value in a fact table that is more flexible is an additive measure. For each dimension you can even sum up. If you want to know the total sales of your company you can easily sum up all the sales.

In below image sales amount we can calculate with all dimensions i.e location wise sales amount, customer wise sales amount, date wise sales amount and product wise sales

Star Schema



Semi-additive

Semi-additive measures can be summed across some dimensions, but not all; checking account or savings account balance amounts are common semi-additive facts.

balance amounts are common semi-additive facts because they are additive across all dimensions except time

ex: my bank balance on 01- aug s 1000

my balance on 01- sep 500

to find total balance we can't sum it bcz total balance available in my account is 500 only not 1500

Semi-Additive:

Semi-additive Facts are Facts that can be summed up for some of the dimensions in the Fact table, but not the others. One of the usual examples for this is are the current account or savings account **balance amounts are common semi-additive facts**. We can get/generate a balance amount from the overall transactions , **but it doesn't make any sense to add (group) the balance amounts from different dates or months or across the time dimension.**

U can aggregate semi-additive facts **for some, but not all, dimensions.**

You can aggregate department headcounts to give an organization total, but you cannot aggregate them over time, so the Sales department headcount for March 31 may be 20 employees, and for April 30 the headcount may be 23, but that does not mean that the total headcount at the end of April is 43.

Non additive measures

Types of Measures/Facts

- Non-Additive Measures/Facts
 - cannot be added across any of the dimensions in the cube. For Eg: Pass Marks etc
- Derived Measures/Facts → (Calculations)
 - Calculated from one or more base facts
 - These are not stored in the cube and are calculated on the fly at the time of accessing them.
 - For example, profit margin

Activate Win
Go to Settings to

Non-Additive:

Certain measures/numbers are completely non-additive, such as ratios discounts passmarks. Non-additive Facts are Facts that cannot be summed up for any of the dimensions present in the Fact table.

Eg: Facts which have percentages, Ratios calculated.

Ex: one employee has sold an item with a 55% profit margin and the other has sold an item with a 45% profit margin, the profit margin for the department is not 100%.

Fk				Non Additive
SubID	Scored	MaxMarks	PassMarks	
S1	67	100	35	
S2	89	100	35 +	
S3	80	100	35	
S4	57	100	35	
	293	400		

In this example passmarks is non additive we can't perform any operations on that.

Derived measures/facts :

Derived Measures/Facts → (Calculations)

- Calculated from one or more base facts
- These are not stored in the cube and are calculated on the fly at the time of accessing them.
- For example, profit margin

In below image profit is not part of db we will calculate on the fly.

Fk			
PID	SoldPrice	PurChasePrice	Profit
1	120	100	20
2	89	55	34
3	109	88	+21

Factless Fact Table

- Factless Measures/Facts
 - Fact table does not have any facts...it has only FKs
 - Eg: Student Attendance

Fact Less Fact Table

- Fact Table which does not contain any measures or facts
- Fact Table contains only key Attributes

Fact Less Fact Table	
DimStudent	
PK	SID
	Name
	Gender

FatctAttendance	
Fk	SID
Fk	CID
Fk	TimeKEY

DimCourse	
PK	CID
	Name
	Fee
	Type

DimTime	
PK	TimeKEY
	Year
	Qtr
	Month
	Week
	Day

A factless fact table is a fact table that does not have any measures.OR facts it contains only key attributes i.e FK

In above image Fact attendance table is factless fact table. From this table we can analyze data like in total strength of given course, how many people are absent in given course and given year, which student has max attendance

Textual Measures:

- Textual Measures/Facts

- Refers to textual data present in the fact table, which is not measurable (non-additive), but is important for analysis purposes. For example, codes (i.e. product codes), flags (i.e. status flag), etc.

In below example bill no is textual measure

Fk				
PID	SoldPrice	PurChasePrice	Profit	Bill Num
1	120	100	20	A-283
2	89	55	34	A-498
3	109	88	21	A-880

Surrogate key:

<https://dwgeek.com/data-warehouse-surrogate-key-design-advantages-disadvantages.html/#:~:text=Data%20warehouse%20surrogate%20keys%20are,join%20dimension%20and%20fact%20tables.&text=It%20is%20just%20sequentially%20generated,better%20lookup%20and%20faster%20joins.>

Data Warehouse Surrogate Keys						
dim_table				fact_table		
PATIENT_SK	PATIENT_ID	PATIENT_NAME	PATIENT_AGE	FCT_SK	AMT	DATE
1	P001	ABC	20	1	1000	1/1/2017
2	P002	BCD	25	2	1500	1/2/2017
3	P003	CDE	19	3	700	1/3/2017
4	P004	DEF	45	4	1200	1/4/2017

DWgeek.com

What are surrogate keys in Data warehouse?

If you are a data warehouse developer, that you might be thinking what is surrogate key? How and where it is being used? You will get answers to all your questions here.

Data warehouse surrogate keys are sequentially generated meaningless numbers associated with each and every record in the data warehouse. These surrogate keys are used to join [dimension](#) and [fact tables](#).

- Usually, [database sequences](#) are used to generate surrogate key so it is always **unique number**
- Surrogate keys **cannot be NULLs**. Surrogate key are never populated with NULL values.
- It does not hold any meaning in data warehouse, often called meaningless numbers. It is just **sequentially generated INTEGER** number for better lookup and faster joins.

Why surrogate keys are used in Data warehouse?

Basically, surrogate key is an artificial key that is used as a substitute for natural key (NK) defined in data warehouse tables. We can use natural key or business keys as a primary key for tables. However, it is not recommended because of following reasons:

- **Natural keys (NK) or Business keys** are generally alphanumeric values that is not suitable for index as traversing become slower. For example, prod123, prod231 etc
- **Business keys are often reused after sometime. It will cause the problem as in data warehouse we maintain historic data as well as current data.**

For example, product codes can be revised and reused after few years. It will become difficult to differentiate current products and historic products. To avoid such a situation, surrogate keys are used.

Data Warehouse Surrogate Key examples

Surrogate Keys are integers that are assigned sequentially in the dimension table which can be used as primary key. The surrogate key column could be identity column or database sequences are used.

Below is the sample example of surrogate key:

PATIENT_SK	PATIENT_ID	PATIENT_NAME	PATIENT_AGE
1	P001	ABC	20
2	P002	BCD	25
3	P003	CDE	19
4	P004	DEF	45

Advantages of Surrogate Key

Below are some of advantages of using surrogate keys in data warehouse:

- With help of surrogate keys, you can integrate heterogeneous data sources to data warehouse if they don't have natural or business keys.
- Joining tables (fact and dimensions) using surrogate key is faster hence better performance
- Surrogate keys are very helpful for ETL transformations.
- Data warehouse Surrogate keys are usually small integer numbers that makes smaller index and better performance
- Surrogate keys are required if you are implementing [slowly changing dimension \(SCD\)](#)

Disadvantages of Surrogate Key

Below are some of disadvantages of using surrogate keys in data warehouse:

- Surrogate key generation and assignment takes unnecessary burden on ETL framework
- You should not over use the surrogate keys as they don't have any meaning in data warehouse tables.
- Data migration becomes difficult if you have database sequence associated with surrogate key columns. You should carefully take care of number surrogate key generation in new database otherwise you may end up with duplicate surrogate keys.

* Surr_Store	Store ID	Street Name	City Name	State Name	Country Name	ZIP Code
1	S1001	New 24 th Blvd	Phoenix	AZ	USA	00000
2	S1002	21 Bell Road	Miami	FL	USA	00000
3	S1003	5 th St Lane	New Port	CA	USA	00000



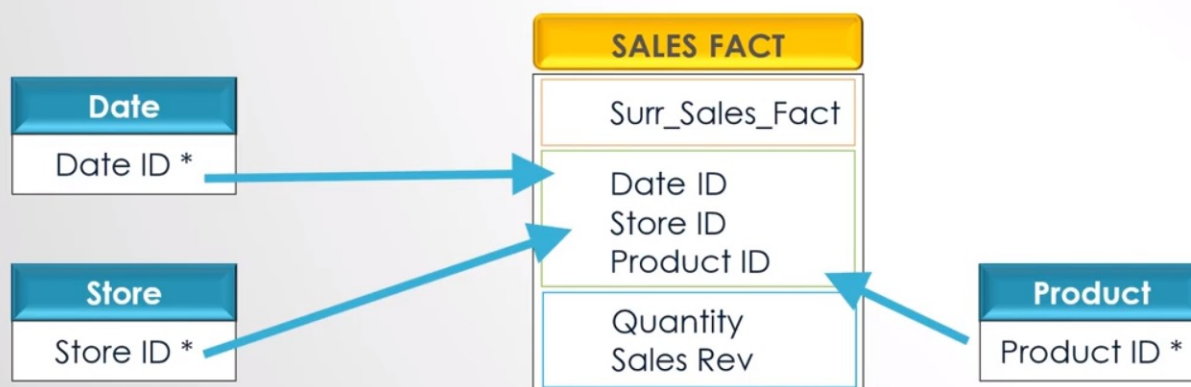
Surrogate Keys

- Anonymous Integer Primary Keys

Grain:



Surr_Sales	Date ID	Store ID	Product ID	Quantity	Sales Rev
1	12/4/2016	Miami	Sweaters	10	5500
2	12/5/2016	Miami	Sweaters	6	4600
3	12/4/2016	Atlanta	Socks	25	650
4	12/5/2016	Atlanta	Socks	30	780





Surr_Sales	Date ID	Store ID	Product ID	Quantity	Sales Rev
1	12/4/2016	Miami	Sweaters	10	5500
2	12/5/2016	Miami	Sweaters	6	4600
3	12/4/2016	Atlanta	Socks	25	650
4	12/5/2016	Atlanta	Socks	30	780

Grain – Level at which facts are measured.

- What is lowest level of information stored in fact table?



Surr_Sales	Date ID	Store ID	Product ID	Quantity	Sales Rev
1	12/4/2016	Miami	Sweaters	10	5500
2	12/5/2016	Miami	Sweaters	6	4600
3	12/4/2016	Atlanta	Socks	25	650
4	12/5/2016	Atlanta	Socks	30	780

"The grain must be declared before choosing dimensions or facts. "

– Ralph Kimball (Data Warehouse Toolkit)



Recommendation – Keep lowest possible grain.

- Better Reporting
- Increases query performance
- Helps in getting summary view

www.aroundBI.com

When you identify the grain, you specify exactly what a fact table record contains. The grain conveys the level of detail that is associated with the fact table measurements. When you identify the grain, you also decide on the level of detail you want to make available in the dimensional model. If more detail is included, the level of granularity is lower. If less detail is included, the level of granularity is higher.

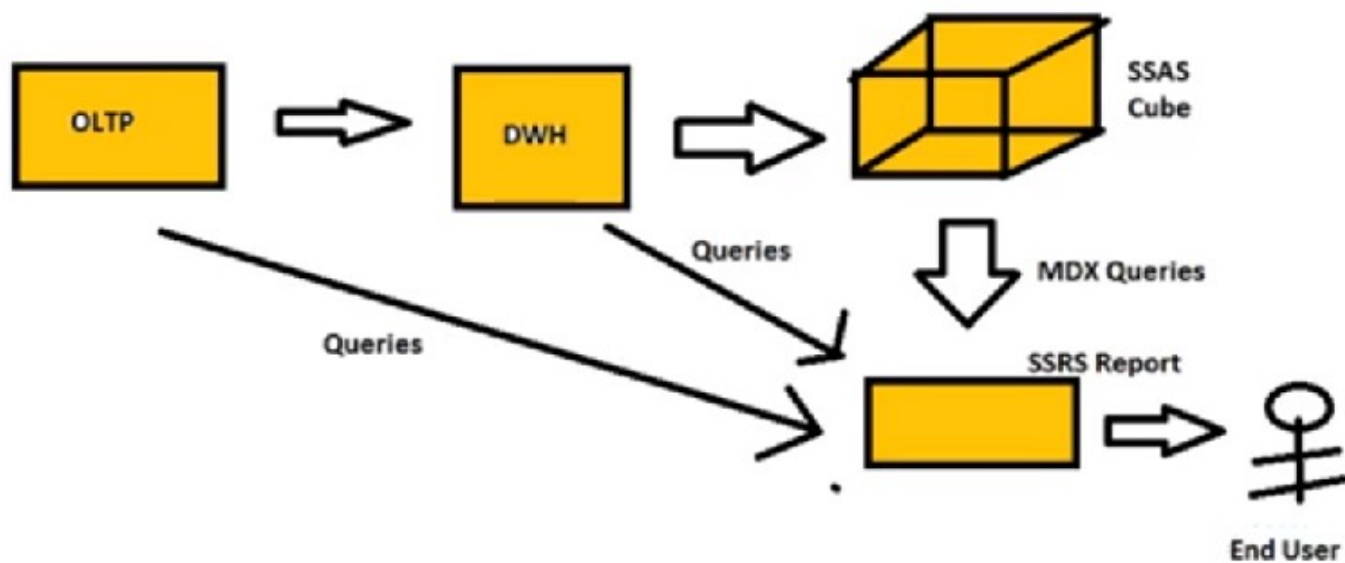
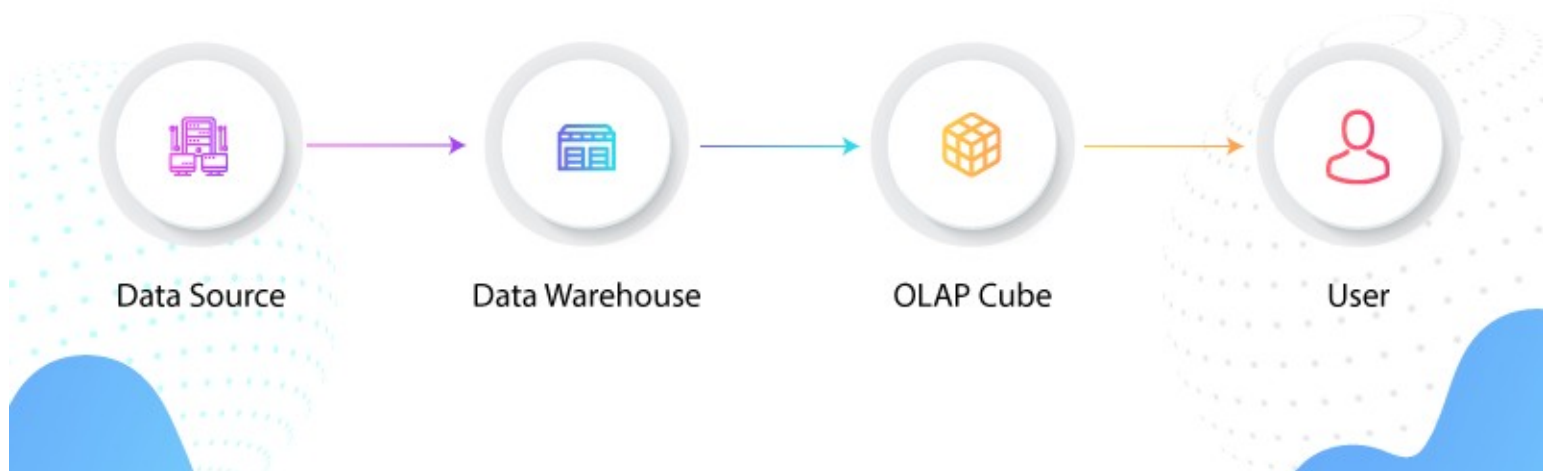
Identifying the level of detail

The level of detail that is available in a star schema is known as the *grain*. Each fact and dimension table has its own grain or granularity. Each table (either fact or dimension) contains some level of detail that is associated with it. The grain of the dimensional model is the finest level of detail that is implied when the fact and dimension tables are joined. For example, the granularity of a dimensional model that consists of the dimensions Date, Store, and Product is *product sold in store by day*.

Data flow

Online Analytical Process (OLAP)

Defined in 200 Words



Aug 2016

Date	QtySupplied	Quality	OnTime
1	2 Ltr	Good	Yes
2	4 Ltr	Bad	Yes
3	1 Ltr	Good	No
4	A	A	A
...			
...			
30	1 Ltr	Good	Yes
31	2 Ltr	Bad	Yes

Aggregations: |

I

Data Warehouse

Date	QtySupplied	quality	OnTime
1	2 Ltr	Good	Yes
2	4 Ltr	Bad	Yes
3	1 Ltr	Good	No
4	A	A	A
...			
...			
30	1 Ltr	Good	Yes
31	2 Ltr	Bad	Yes

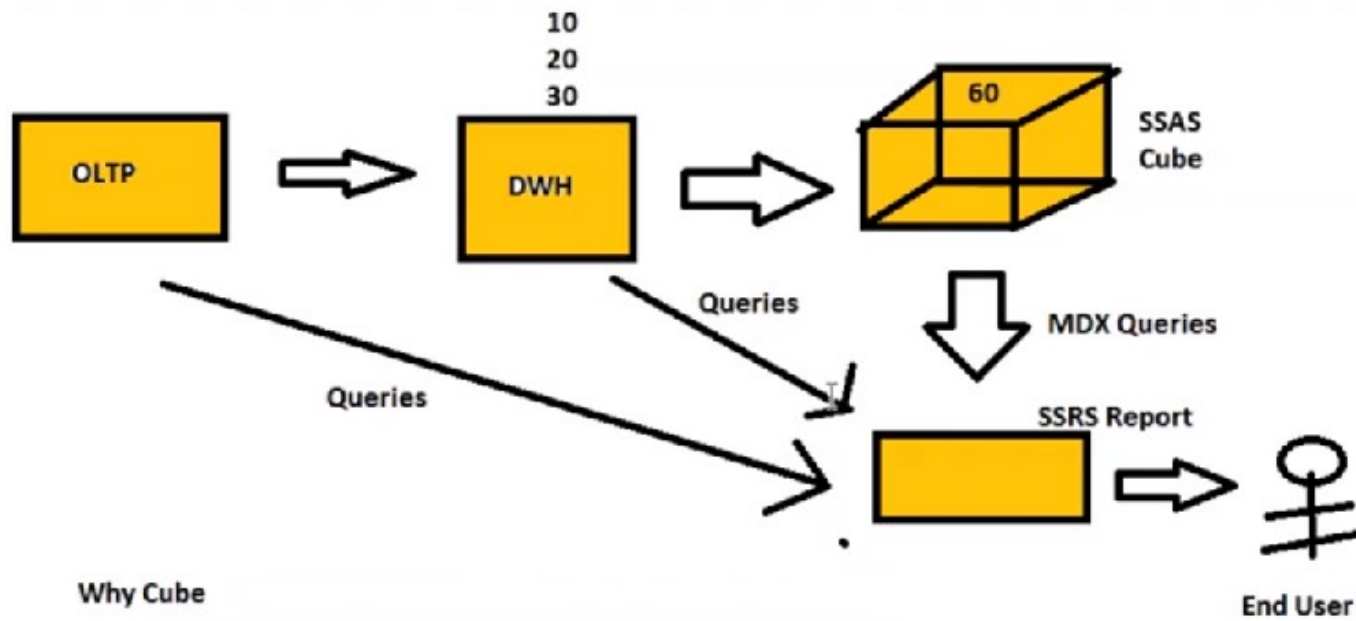
Cube (Aggregated)

Total

Store 4123 -- Aug 2016

Aggregations: $44.5 \times 77 = 4123/=$

. if we write query directly from DWH the query performance is slow. cube will perform all permutations and combinations of data aggregations so we will get o/p very fast



Why Cube

Super Slow BI Application will become super fast BI App

Intro to SSAS

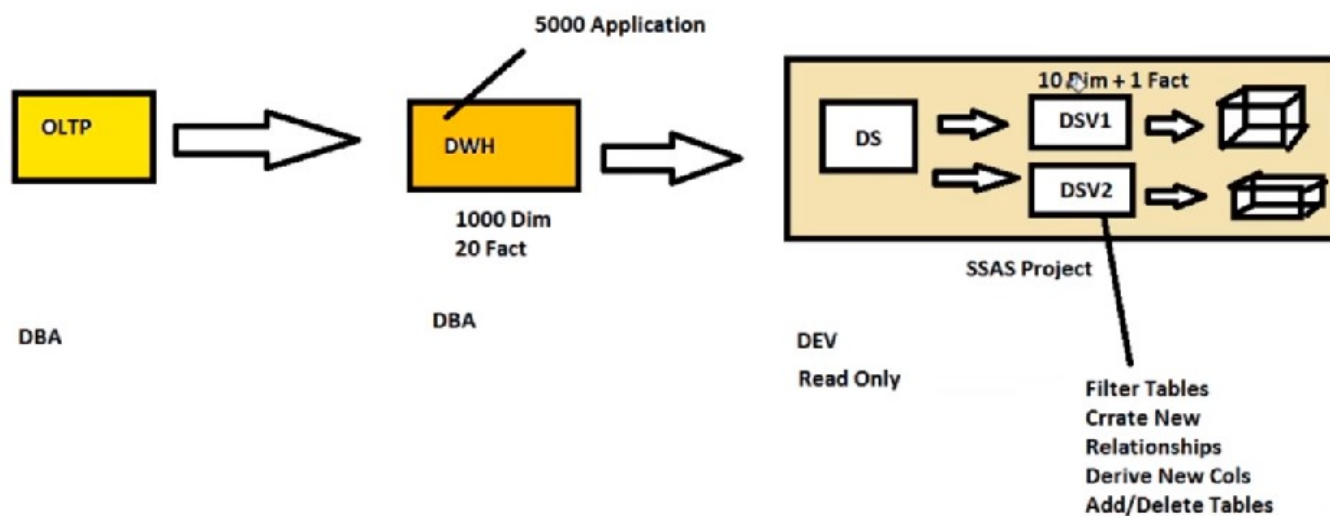
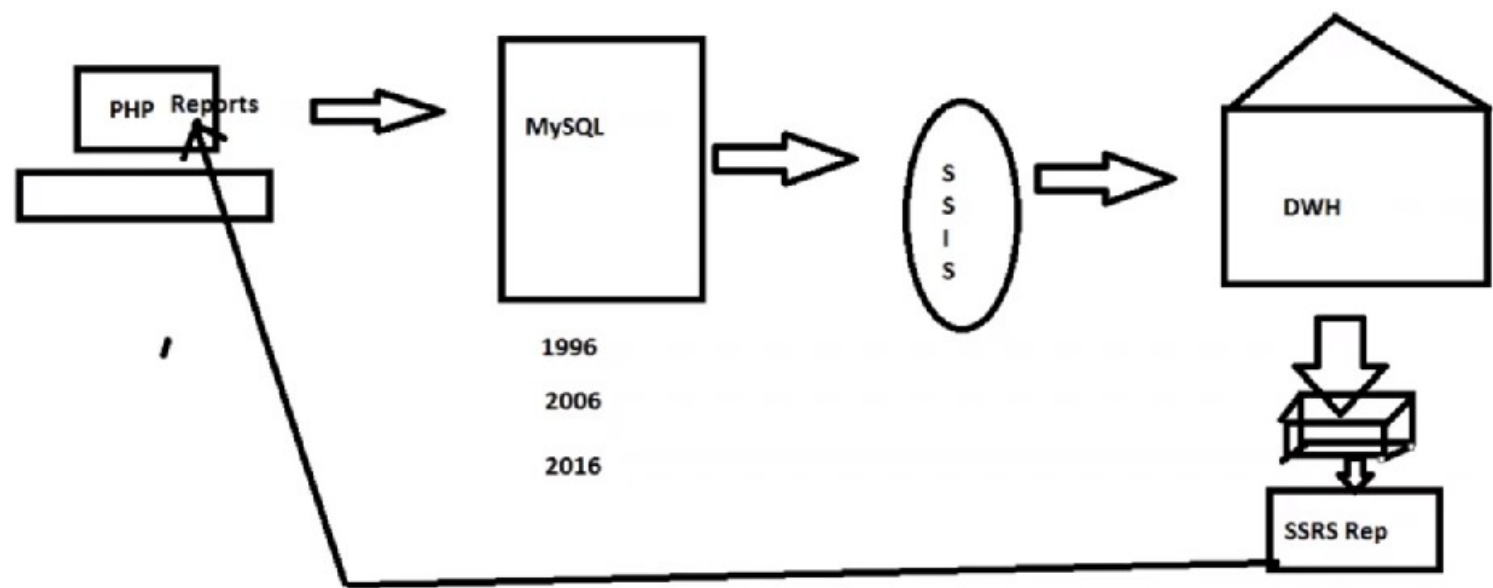
- Is an Analysis Tool
- Stores Multidimensional Aggregated Data to speed up the queries
- Used for Large Data to Aggregate and stores the aggregated data
- Is a Multi Dimensional OLAP Server and Data Mining Server

Roles of SSAS

- OLAP Server
 - To store Pre-aggregate Values (Totals) for Multi Dimensional Analysis
 - Understand the Past
- Data Mining Server
 - For Understand Trends and Patterns
 - For Predictions
 - Understand the Future

Need for SSAS

- SSAS makes super slow BI application to super fast BI application, by using pre-aggregated totals for multi-dimensional analysis
- Develop BI app using OLTP → Reporting (Or)
- Develop BI app using OLTP → OLAP → Reporting
- Note: Hence SSAS is **NOT** a mandatory component to develop a BI application



olap

OLAP works as a fundamental stone to many types of business apps pertaining to

- Business Performance Management

- Planning, Forecasting, Budgeting
- Financial Reporting
- Simulation Models
- Knowledge Discovery
- Data Warehouse Reporting

Features

- Cube Design
- MOLAP, ROLAP (Multidimensional and Relational OLAP)
- MDX Queries
- Slice and Dice
- Drill down
- Ad-hoc Analysis

Major Advantages

- In-depth analysis because of multiple dimensions
- Empowers complicated analysis of bulky data volumes as against intricate business-queries

Cubes

Cubes are data processing units composed of fact tables and dimensions

from the data warehouse. They provide multidimensional views of data, querying and analytical capabilities to clients. A cube can be stored on a single analysis server and then defined as a linked cube on other Analysis servers. End users connected to any of these analysis servers can then access the cube. This arrangement avoids the more costly alternative of storing and maintaining copies of a cube on multiple analysis servers. linked cubes can be connected using TCP/IP or HTTP. To end users a linked cube looks like a regular cube. Linked cube are cubes in which a sub-set of the data can be analysed into great detail. The linking ensures that the data in the cubes remain consistent.

<https://intellipaat.com/blog/tutorial/data-warehouse-tutorial/what-is-olap-and-multidimensional-model/>

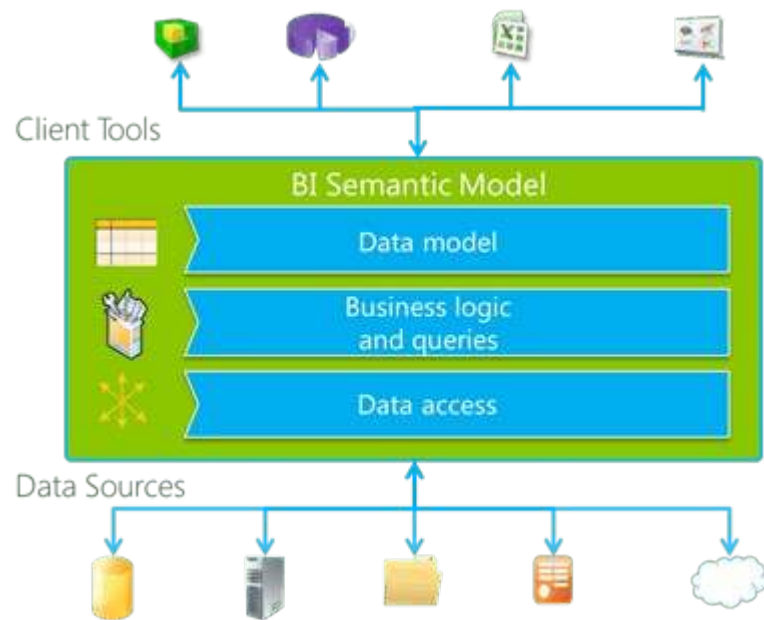
Peeling back 3 layers of BI and Analytics software

by [Ken Gnazdowsky, B.Sc.](#) | Posted on [October 17, 2018](#)

Three inter-related layers or processes in a robust BI analytics solution can include:

1. a Symantic (meta-data) layer;
2. OLAP cubes;
3. Extract, Transform, Load (ETL) jobs.

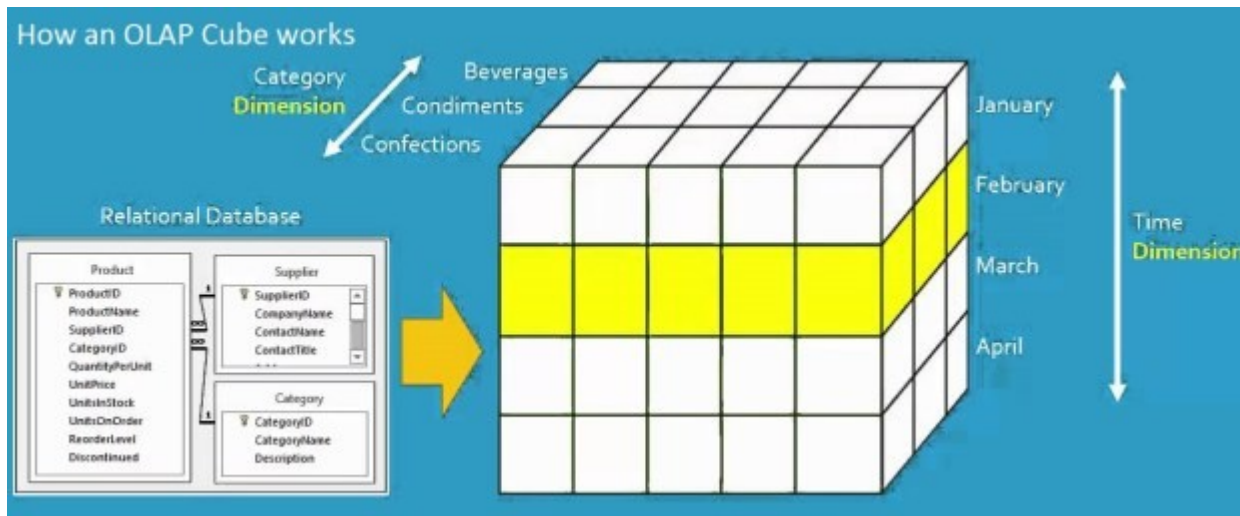
Let's take a closer look at each of these and how small changes within any one of them can affect your critical business reports at the top of your BI stack.



(1) **Semantic layer** – an abstracted (meta-data) layer that contains simplified business views mapped to underlying relational database models and data warehouses. This layer is a pre-requisite for some WSYWIG, drag and drop design reporting tools.

A **semantic layer** is a business representation of corporate data that helps non-technical end users access data using common business terms. The semantic layer is configured by a person who has knowledge of both the data store and the reporting needs of the business. The person creating the semantic layer chooses to expose appropriate fields in the data store as “business fields,” and to hide any fields that are not relevant. Each business field is given a friendly, meaningful name, and the business fields are organized in a way that will make sense to business users.

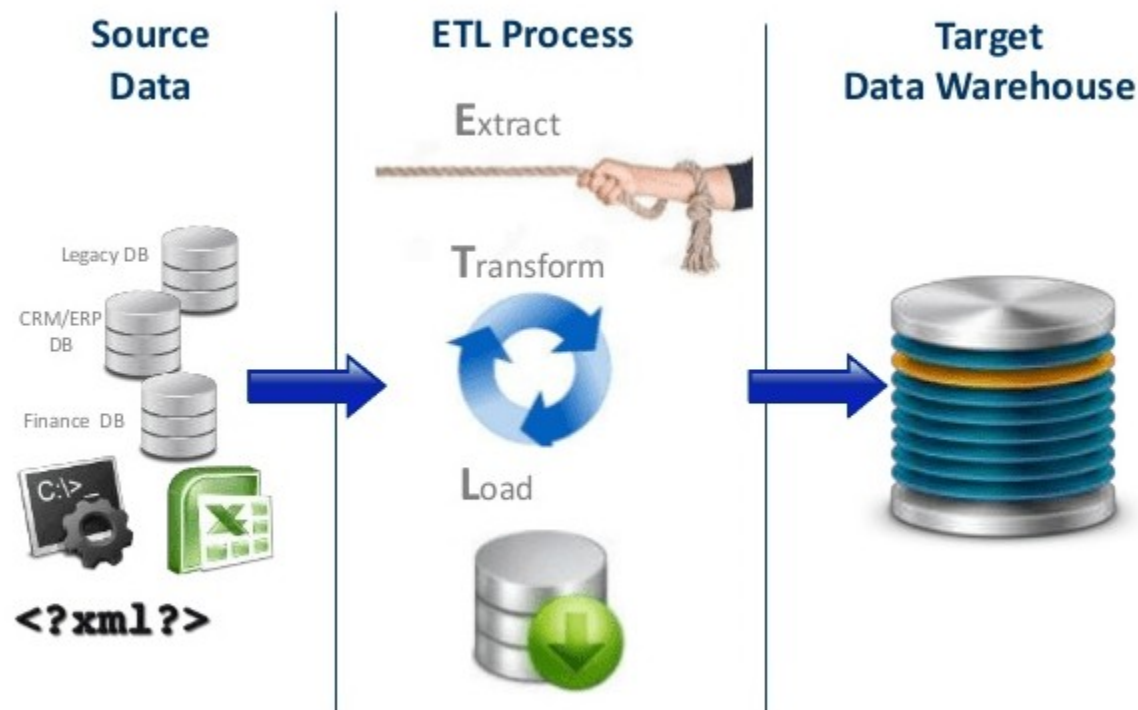
By using common business terms, rather than data language, to access, manipulate, and organize information, a semantic layer simplifies the complexity of business data. This is claimed to be core business intelligence (BI) technology that frees users from IT while ensuring correct results when creating and analysing their own reports & dashboards. In reality, however, *IT staff are typically needed to create and maintain the **semantic layer** (business view definitions) which maps tables to classes and columns to objects.*



(2) **OLAP cubes** – the arrangement of data into Cubes overcomes a limitation of relational databases, which are not well suited for near instantaneous analysis and display of large amounts of data. OLAP data is typically stored in a star schema or snowflake schema within a relational data warehouse or in a special-purpose data management system.

An **OLAP cube** is a term that typically refers to multi-dimensional array of data. *OLAP* is an acronym for online analytical processing, which is a computer-based technique of analyzing data to look for insights. A cube can be considered a multi-dimensional generalization of a two- or three-dimensional spreadsheet. Each cell of the cube holds a number that represents some *measure* of the business, such as sales, profits, expenses, budget and forecast. A Slice represents two-dimensional view of an **OLAP Cube** that arranges data in a grid, similar to a spreadsheet; a Slice functions much like a report or a query in an **RDBMS** in that it returns data based on a request for what to see

Although many report-writing tools exist for relational databases, these are slow when the whole database must be summarized, and present great difficulties when users wish to re-orient reports or analyses according to different, multidimensional perspectives, aka, **Slices**. The use of Cubes facilitate this kind of fast end-user interaction with data. Some reporting tools like Crystal Reports include the ability to define simple two-dimensional OLAP cubes for generating graphs for example.



(3) **ETL Software** – import clean data from virtually any source into permanent, structured relational database tables.

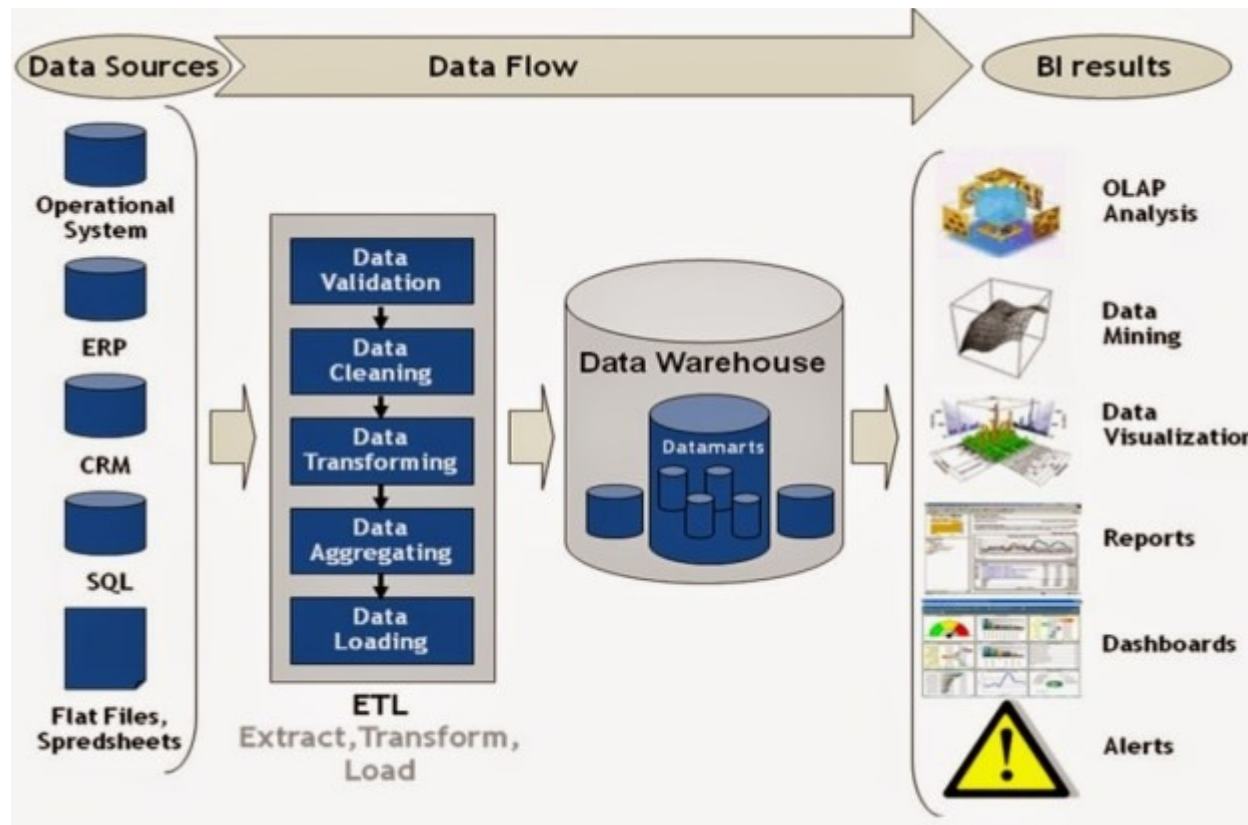
ALSO CALLED: Data Extraction Software, Database Extraction Software

DEFINITION: In managing databases, ETL refers to three separate functions combined into a single programming tool: Extract, Transform, Load

1. First, the **extract** function reads data from a specified source database and extracts a desired subset of data.
2. Next, the **transform** function works with the acquired data – using rules or lookup tables, or creating combinations with other data – to convert it to the desired state.
3. Finally, the **load** function is used to write the resulting data (either all of the subset or just the changes) to a target database, which may or may not previously exist.

***ETL** can be used to acquire a temporary subset of data from virtually any file for reports or other purposes, or a more permanent data set may be acquired for other purposes such as: the*

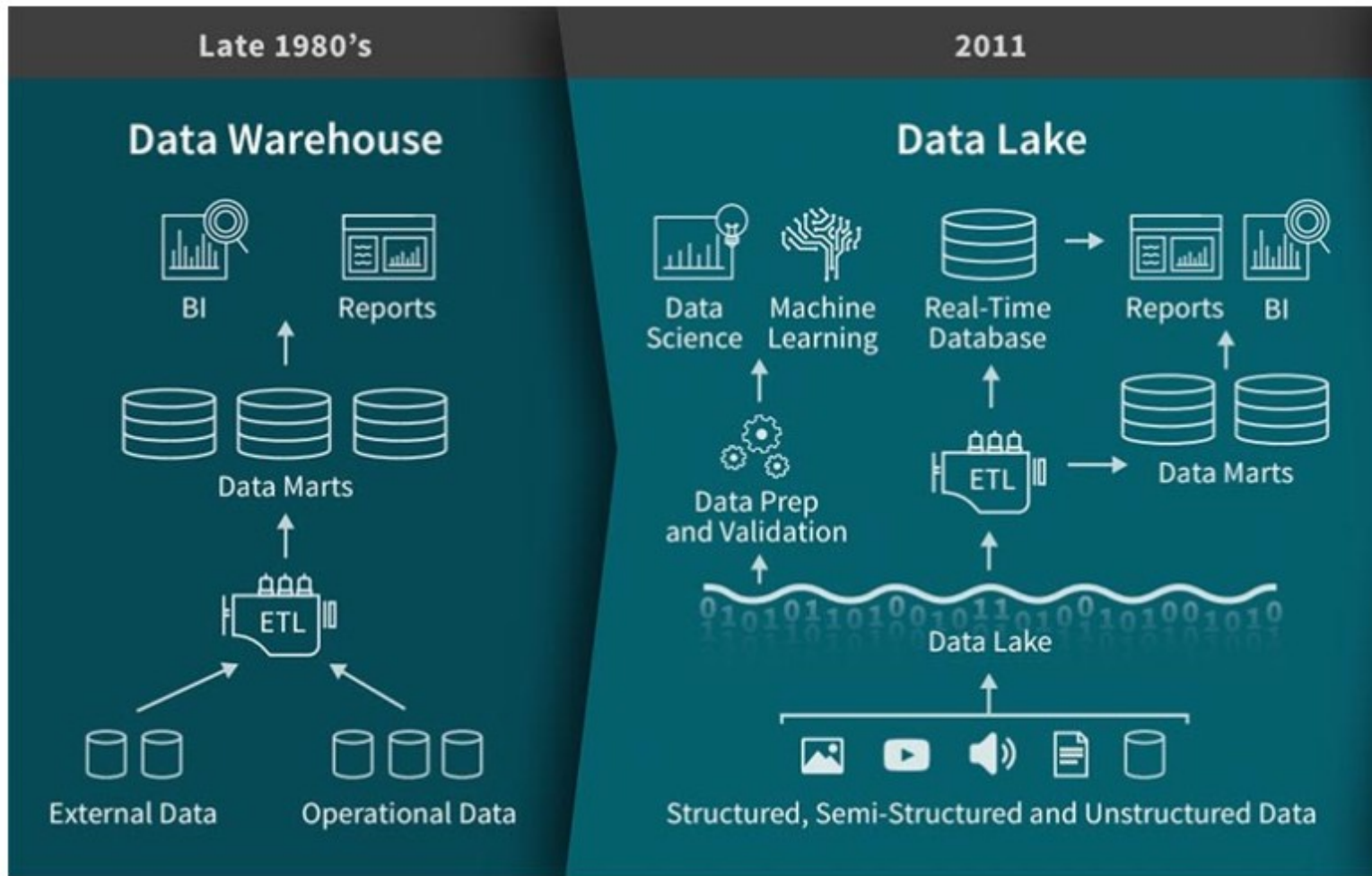
population of a data mart or data warehouse; conversion from one database type to another; and the migration of data from one database or platform to another.



A change to the underlying source database, or at any one of these BI layers can affect the hundreds or in some organizations, thousands of reports that are dependent on the underlying data model. *Something as simple as renaming a field or variable name can essentially break the chain, and stop reports from working or producing accurate results the business depends on.*

Now imagine a change impact analysis tool that can search your entire BI process flow to show you all affected references throughout your Business Intelligence stack.

Data Lake architecture



Because data that goes into data warehouses needs to go through a strict governance process before it gets stored, adding new data elements to a data warehouse means changing the design, implementing or refactoring structured storage for the data and the corresponding ETL to load the data. With a massive amount of data, this process could require significant time and resources. This is where a data lake concept comes into the picture and becomes a game-changer in big data management.

The concept of data lake emerges in the 2010s, which, in a simple language, is the idea that all enterprise's structured, unstructured and semi-structured data can and should be stored in the same place. Apache Hadoop is an example of data infrastructure that allows to store and process massive amounts of data, both structured and unstructured; which enables the Data Lake architecture.

business questions. However, the biggest concern with data lake is that, without appropriate governance, data lakes can quickly turn into unmanageable data swamps. **Put it differently, without knowing how the water is in a lake, who would want to go swim in it? Business users can't not utilize the data lake if they don't trust the data quality of that lake .**



Source; Datavercity through [slideshare](#)

Recently the trend of companies that want to benefit from a data lake architecture in a more conservative way has emerged. These companies are stepping away from the ungoverned “free-entry” approach and instead developing a more governed data lake.

The data lake can contain two environments: an exploration/development and production environment. Data will be explored, cleansed, transformed in order to build machine learning model, build functions and other analytics purposes. Data such as metrics, functions that have been generated by the transformation process will be stored in the production part of data lake.

Another trend is, rather than pouring all raw data into the lake, the governed data lake only allows ‘verified’ data to get into it. Essentially, a governed data lake architecture does not restrict the types of data that are stored in it, meaning that governed data lakes still comprise multiple data types including unstructured and semi-structured data like XML, JSON, CSV. However, the key is to make sure that no data is stored in the lake without being described and documented in business glossary, which will give some confidence to the users about the quality and meaning of data.

To provide this layer of governance, a business glossary tool has to be in place to document the meaning of the data. More importantly, there needs to be a governance process around this — which is all about roles and responsibilities, for instance, who owns the data, who defines it, who will be responsible for any data quality issues. Going for this approach will be time-consuming because defining data itself can be a long process since it involves people from different disciplines across an enterprise.

Data Lake	Data warehouse
Schema on read	Schema on write.
Can store structure, unstructured and semi-structured data	Mainly store structure data
Is optimized for cost efficiency	Is optimized for reporting purpose.
Store raw data	Store processed and governed data
ELT process	ETL process

Data Lake and Data Warehouse comparison. Illustration by author.

Defining Identifying and Non-Identifying Relationships

In database terms, relationships between two entities may be classified as being either identifying or non-identifying. Identifying relationships exist when the primary key of the parent entity is included in the primary key of the child entity. On the other hand, a non-identifying relationship exists when the primary key of the parent entity is included in the child entity but not as part of the child entity's primary key. In addition, non-identifying relationships may be further classified as being either mandatory or optional. A “mandatory” non-identifying relationship exists when the value in the child table cannot be null. On the other hand, an “optional” non-identifying relationship exists when the value in the child table can be null.